

Jack L. Gallant, Ph.D.

Curriculum Vitae

May, 2026

Current Position

Professor and Class of 1940 Chair
Department of Neuroscience
co-Director, UC Berkeley Henry J. Wheeler Brain Imaging Center
Courtesy Appointment in Department of Electrical Engineering and Computer Science
Member, Graduate programs in Bioengineering, Biophysics, Vision Science
Member, Institute of Brain and Cognitive Sciences
University of California at Berkeley

Research Interests and Personal Statement

The overall goal of my research program is to understand and model how information is represented in the brain under natural conditions (i.e., natural stimulation and natural tasks). Until the last few years most of my work focused on vision. I addressed two broad questions in that work: How is information about the visual world represented across the brain? And how are these representations modulated by attention, learning and memory? I used three convergent methods to investigate these issues: neurophysiology, functional MRI, and computational modeling. When I was a post-doc in David Van Essen's lab I first discovered that area V4 represents non-Cartesian features such as curvature and intersections (Gallant et al. 1993; 1996). However, it was clear to me at that time that the representations in V4 were far more complicated than what could be revealed with simple stimuli. I therefore began to use natural images as stimuli in order to attack the problem more directly (Gallant et al 1998). After moving to UC Berkeley in 1995 I continued to investigate the neural mechanisms mediating natural vision (Vinje & Gallant 2000; David et al 2004; Nishimoto et al 2011; Willmore et al 2011) and attention (Mazer & Gallant 2003; Hayden et al 2006; David et al 2008; Hayden & Gallant, 2013). I also developed new statistical and computational tools for modeling neuronal responses to complex stimuli (Theunissen et al 2001; Prenger et al 2004; David et al 2004; David et al 2006; Wu et al 2006). After Berkeley installed an MRI machine in 2002 I began to adapt the computational modeling algorithms that I had originally developed for neurophysiology so that they could be applied to functional MRI (Hansen et al 2004; Huth et al 2012; Kay et al 2008; Naselaris et al 2009; Nishimoto et al 2011). I also worked to optimize MRI data acquisition (Hansen et al 2007; Kay et al 2008), pre-processing (Kay et al 2008) and visualization (Gao et al 2015). This line of work produced a new voxelwise modeling approach that has proven to be the most sensitive method available for analyzing fMRI data and for decoding brain activity (Kay et al 2008; Naselaris et al 2009; Nishimoto et al 2011; Huth et al 2012; Cukur et al 2013a, b; Stansbury et al 2013; Huth et al 2016). My laboratory currently focuses on using fMRI, statistical and computational modeling to produce detailed human cortical functional maps of information related to vision, language and decision making; systematizing human functional anatomy; characterizing individual differences in cortical organization in humans; understanding dynamic thought processes in the human brain; and decoding human brain activity. We also have an active program of research and collaboration aimed at translating our powerful functional imaging and computational modeling methods for use in the diagnosis, assessment and monitoring of mental, neurological and developmental brain diseases and disorders.

Education

1988: Ph.D., Yale University
1983: B.A., University of California at Berkeley

Previous Positions

2002-2008: Associate Professor, UC Berkeley
1995-2002: Assistant Professor, UC Berkeley
1992-1995: Post-doctoral researcher, Washington University Medical School
1988-1992: Post-doctoral researcher, California Institute of Technology

Honors

2009-2025: Keynote or plenary speaker at 20 scientific conferences
2025: Elected as AAAS Fellow
2022: Elected as Chair, IEEE Brain Technical Community (one year position)
2020: Elected as a Fellow of the Association for Psychological Science
2017: Appointed as Class of 1940 Chair at UC Berkeley
2016: Elected as Fellow of the Society of Experimental Psychologists
2015: Elected as Senior Member of IEEE
2015: Appointed as Chancellor's Professorship at UC Berkeley
2011: One of Time Magazine's 50 best inventions of the year

2000: Hellman Fellow, UC Berkeley
1998: Sloan Foundation Fellowship
1997: Whitehall Foundation Award

Most Impactful Publications (Science, Nature, and those with more than 1000 citations)

- Huth, A. G., de Heer, W. A., Griffiths, T. L., Theunissen, F. E., & **Gallant, J. L.** (2016). Natural speech reveals the semantic maps that tile human cerebral cortex. *Nature*, 532(7600), 453-458. [NIHMSID: 765514, Publ.ID: 2014-01-00140, DOI: <http://dx.doi.org/10.1038/nature17637>]
- Huth, A.G., Nishimoto, S., Vu, A., & **Gallant, J.L.** (2012). A continuous semantic space describes the representation of thousands of object and action categories across the human brain. *Neuron*, 76, 1210-1224. [DOI: 10.1016/j.neuron.2012.10.014, PMCID: PMC3556488, NIHMSID: NIHMS418681]
- Nishimoto, S., Vu, A.T., Naselaris, T., Benjamini, Y., Yu, B., & **Gallant, J.L.** (2011). Reconstructing visual experiences from brain activity evoked by natural movies. *Current Biology*, 21, 1641-1646. [DOI: 10.1016/j.cub.2011.08.031, PMID: 21945275]
- Naselaris, T., Kay, K.N., Nishimoto, S., & **Gallant, J.L.** (2011). Encoding and decoding in fMRI. *Neuroimage*, 52(2), 400-410.
- Kay, K.N., Naselaris, T., Prenger, R.J., & **Gallant, J.L.** (2008). Identifying natural images from human brain activity. *Nature*, 452, 352-355.
- Vinje, W.E. & **Gallant, J.L.** (2000). Sparse coding and decorrelation in primary visual cortex during natural vision. *Science*, 287, 1273-1276.
- Gallant, J.L.**, Braun, J. & Van Essen, D.C. (1993). Selectivity for polar, hyperbolic, and Cartesian gratings in macaque visual cortex. *Science*, 259, 100-104.

Other Selected Publications

- Zeng, A., & **Gallant, J.L.** (2025). Disentangling Superpositions: Interpretable Brain Encoding Model with Sparse Concept Atoms. *NeurIPS*. [<https://openreview.net/pdf?id=3aNvX9TQTo>]
- Dixon, T.C., Strandquist, G., Zheng, A., Fraczek, T., Bechtold, R., Lawrence, D., Ravi, S., Starr, P.A., Gallant, J.L., & Herron, J.A., Little, S.J. (2025). Movement-responsive deep brain stimulation for Parkinson's Disease using a remotely optimized neural decoder. *Nature Biomedical Engineering*, 1-15.
- Dupré la Tour, T., Visconti di Oleggio Castello, M. & **Gallant, J.L.** (2025). The voxelwise modeling framework: a tutorial introduction to fitting encoding models to fMRI data. *Imaging Neuroscience*, 3, imag_a_00575.
- Chen, C., Dupré la Tour, T., Gallant, J. L., Klein, D., & Deniz, F. (2024). The cortical representation of language timescales is shared between reading and listening. *Communications Biology*, 7(1), 284. [PMCID: PMC11245628]
- Abdel-Ghaffar, S., Huth, A., Lescroart, M., **Gallant, J.L.**, Stansbury, D., Bishop, S. (2024). Occipital-temporal cortical tuning to semantic and affective features of natural images predicts associated behavioral responses. *Nature Communications*, 15(531), 1-17. [PMCID: PMC11233618]
- Gong, X.L., Huth, A.G., Deniz, F., Johnson, K., **Gallant, J.L.**, Theunissen, F.E. (2023). Phonemic segmentation of narrative speech in human cerebral cortex. *Nature Communications*, 14, 1-17. [PMCID: PMC10354060]
- Deniz, F., Tseng, C., Wehbe, L., & **Gallant, J.L.** (2023). Semantic representations during language comprehension are affected by context. *J. Neurosci*, 43(17), 3144-3158. [PMCID: PMC10146529]
- Dupre la Tour, T., Eickenberg, M., Nunez-Elizalde, A.O., & **Gallant, J.L.** (2022). Feature-space selection with banded ridge regression. *NeuroImage*, 264, 119728. [PMCID: PMC9807218]
- Popham, S.F., Huth, A.G., Bilenko, N.Y., Gao, J.S., Nunez-Elizalde, A.O., & **Gallant, J.L.** (2021). Visual and linguistic semantic representations are aligned at the border of human visual cortex. *Nature Neuroscience*, 24, 1628–1636. [PMID: 34711960]
- Goering, S., Klein, E., Sullivan, L. S., Wexler, A., y Arcas, B. A., Bi, G., ... **Gallant, J.L.**, ... & Yuste, R. (2021). Recommendations for Responsible Development and Application of Neurotechnologies. *Neuroethics*, 1-22. [PMCID: PMC8081770]
- Gallant, J.L.** & Popham, S. (2020). Semantic representation in the human brain under rich, naturalistic conditions. *The Cognitive Neurosciences*, Gazzaniga, Mangun, Poeppel (Eds.), MIT Press.
- Zhang, T., Gao, J.S., Cukur, T., & **Gallant, J.L.** (2020). Voxel-based state space modeling recovers task-related cognitive states in naturalistic fMRI experiments. *Frontiers in Neuroscience*, 14:565976. [PMID: 34045937]
- Deniz, F., Nunez-Elizalde, A.O., Huth, A.G., & **Gallant, J.L.** (2019). The representation of semantic information across human cerebral cortex during listening versus reading is invariant to stimulus modality. *Journal of Neuroscience*, 39(39), 7722-7736. [PMCID: PMC6764208]
- Nunez-Elizalde, A.O., Huth, A.G., & **Gallant, J.L.** (2019). Voxelwise encoding models with non-spherical multivariate normal priors. *NeuroImage*, 197, 482-492. [PMID: 31075394]
- Lescroart, M.D., & **Gallant, J.L.** (2019). Human scene-selective areas represent 3D configurations of surfaces. *Neuron*, 101, 178-192. [PMID: 30497771]
- de Heer, W.D., Huth, A.G., Griffiths, T.L., **Gallant, J.L.**, & Theunissen, F.E. (2017). The hierarchical cortical organization of human speech processing. *Journal of Neuroscience*, 37:27, 6539-6557. [PMCID: PMC5511884]
- Cukur, T.C., Huth, A.G., Nishimoto, S., & **Gallant, J.L.** (2016). Functional subdomains within scene-responsive ROIs: Parahippocampal Place Area, Retrosplenial Complex, and Occipital Place Area. *Journal of Neuroscience*, 36:40, 10257-10273.

[PMCID: PMC5050324]

- Huth, A.G., Lee, T., Nishimoto, S., Bilenko, N.Y., Vu, A.T., & **Gallant, J.L.** (2016). Decoding the semantic content of natural movies from human brain activity. *Frontiers in Systems Neuroscience*, 10:81. [PMCID: PMC5057448]
- Lescroart, M.D., Stansbury, D., & **Gallant, J.L.** (2015). Fourier power, subjective distance and semantic categories all provide plausible models of BOLD responses in scene-selective visual areas. *Frontiers in Computational Neuroscience*, 9:135. [PMCID: PMC4633487]
- Gao, J.S., Huth, A.G., Lescroart, M.D., & **Gallant, J.L.** (2015). Pycortex: an interactive surface visualizer for fMRI. *PLOS Neuroinformatics*, 9:23. [PMCID: PMC4586273]
- Cukur, T.C., Huth, A.G., Nishimoto, S., & **Gallant, J.L.** (2013). Functional subdomains within human FFA. *Journal of Neuroscience*, 33:42, 16748-16766. [DOI: 10.1523/JNEUROSCI.1259-13.2013]
- Stansbury, D.E., Naselaris, T., Vu, A.T., & **Gallant, J.L.** (2013). Natural scene statistics explain the representation of scene categories in human visual cortex. *Neuron*, 79, 1025-1034.
- Cukur, T., Nishimoto, S., & **Gallant, J.L.** (2013). Attention During Natural Vision Warps Semantic Representation Across the Human Brain. *Nature Neuroscience*, 16:6, 763-770. [DOI: 10.1038/nn.3381, PMID: 23603707, NIHMSID: 457964]
- Hayden, B.Y. & **Gallant, J.L.** (2013). Working memory and decision processes in visual area V4. *Frontiers in Neuroscience*, 7(18), 1-10. [DOI: 10.3389/fnins.2013.00018, PMID: 23550043]
- Nishimoto, S., & **Gallant, J.L.** (2011). A three-dimensional spectral receptive field model explains responses of area MT neurons to natural movies. *Journal of Neuroscience*, 31, 14551-14564. [DOI: 10.1523/JNEUROSCI.6801-10.2011, PMID: 21994372]
- David, S.V., Hayden, B.J., Mazer, J. A., & **Gallant, J.L.** (2008). Attention to stimulus features shifts spectral tuning of V4 neurons during natural vision. *Neuron*, 59, 509-521. [DOI: 10.1016/j.neuron.2008.07.001, PMCID: PMC2948549]
- Hansen, K.A., Kay, K.N. & **Gallant, J.L.** (2007). Topographic organization in and near human visual area V4. *Journal of Neuroscience*, 27, 11896-11911.
- Wu, M.C.-K., David, S. V., & **Gallant, J.L.** (2006). Complete functional characterization of sensory neurons by system identification. *Annual Review of Neuroscience*. 29, 477-505.
- Hayden, B.Y. & **Gallant, J.L.** (2005). Timecourse of attention reveals different mechanisms for spatial and feature attention in area V4. *Neuron*, 47, 637-643.
- Hansen, K.A., David, S.V. & **Gallant, J.L.** (2004). Parametric reverse correlation reveals spatial linearity of retinotopic human V1 BOLD response. *Neuroimage*, 23, 233-241.
- David, S.V., Vinje, W.E. & **Gallant, J.L.** (2004). Natural stimulus statistics alter the receptive field structure of V1 neurons. *Journal of Neuroscience*, 24, 6991-7006.
- Mazer, J.A. & **Gallant, J.L.** (2003). Goal-related activity in area V4 during free viewing visual search: Evidence for a ventral stream salience map. *Neuron*, 40, 1241-1250.
- Vinje, W.E. & **Gallant, J.L.** (2002). Natural stimulation of the non-classical receptive field increases information transmission efficiency in V1. *Journal of Neuroscience*, 22, 2904-2915.
- Mazer, J.A., Vinje, W.E., McDermott, J., Schiller, P.H. & **Gallant, J.L.** (2002). Efficient characterization of spatial frequency and orientation tuning dynamics in area V1 of awake behaving animals. *Proceedings of the National Academy of Sciences USA*, 99, 1645-1650.
- Gallant, J.L.**, Shoup, R.E. & Mazer, J.A. (2000). A human extrastriate cortical area that is functionally homologous to Macaque area V4. *Neuron*, 27, 227-235.
- Gallant, J.L.**, Connor, C.E. & Van Essen, D.C. (1998). Neural activity in areas V1, V2 and V4 during free viewing of natural scenes compared to controlled viewing. *NeuroReport*, 9, 2153-2158.
- Connor, C.E., Preddie, D.C., **Gallant, J.L.** & Van Essen, D.C. (1997). Spatial attention effects in macaque area V4. *Journal of Neuroscience*, 17, 3201-3214.
- Gallant, J.L.**, Connor, C.E., Rakshit, S., Lewis, J.L. & Van Essen, D.C. (1996). Neural responses to polar, hyperbolic, and Cartesian gratings in area V4 of the macaque monkey. *Journal of Neurophysiology*, 76, 2718-2739.